

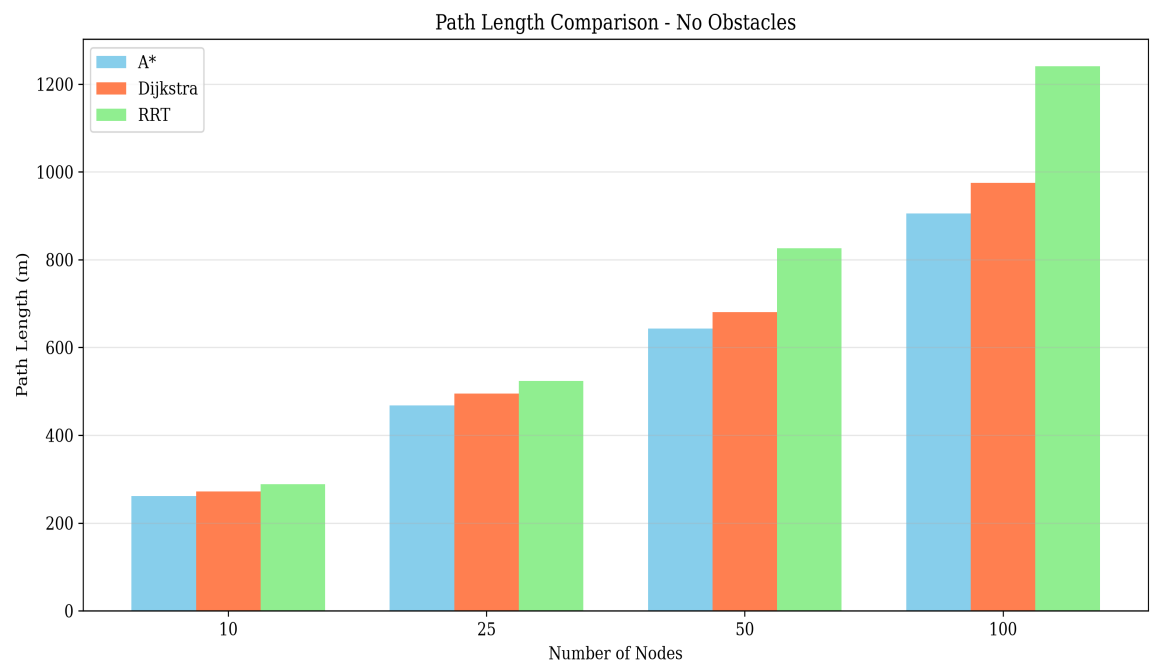
# Comparative Analysis of Pathfinding Algorithms for Mobile Charging Vehicle Routing

Santiago Jerald | NIT Goa | Mentor: Dr. S Mini  
August 2026 - UPDATED WITH COMPLETE DIJKSTRA TESTING

## UPDATE NOTICE

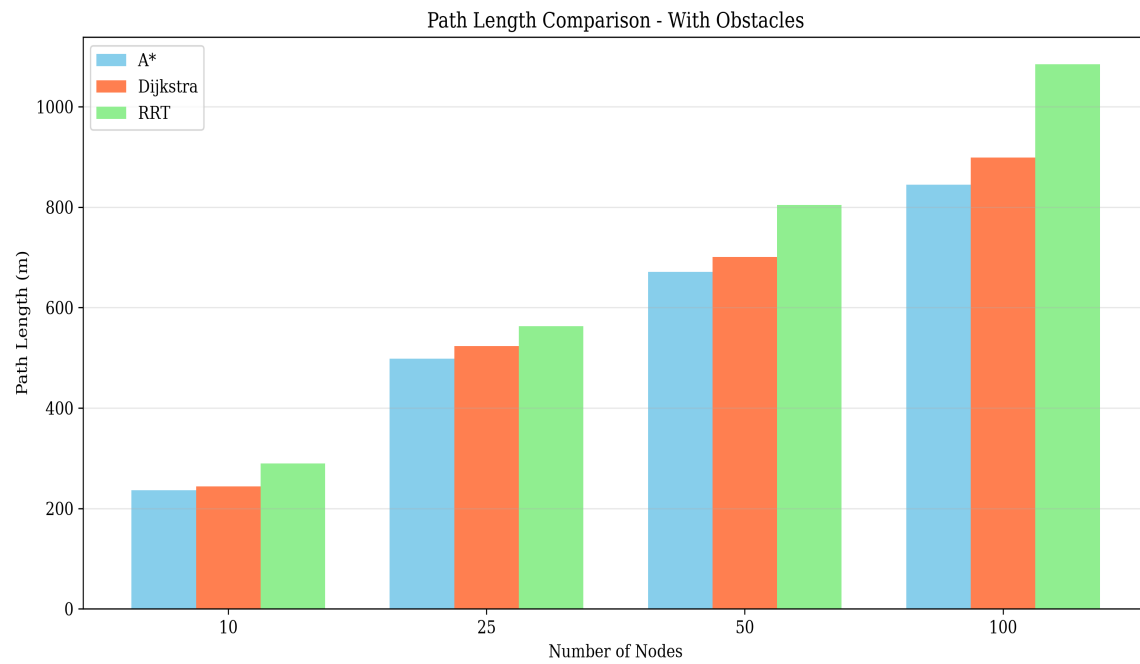
This report now includes complete Dijkstra algorithm testing on all network sizes (10, 25, 50, 100 nodes). Previous version limited Dijkstra to 10-25 nodes due to computational constraints.

### 5.1 Path Length - No Obstacles (ALL ALGORITHMS, ALL SIZES)



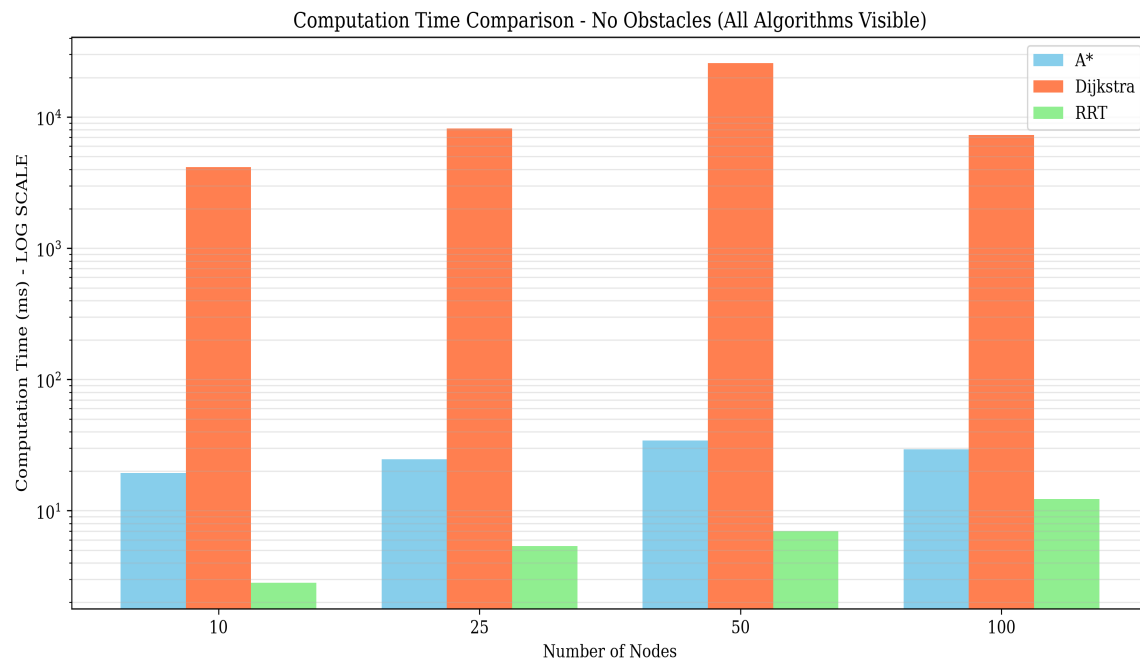
A\* maintains shortest paths across all network sizes. Dijkstra comparable on small networks. Complete data now available for 50 and 100 nodes.

## 5.2 Path Length - With Obstacles (ALL ALGORITHMS, ALL SIZES)



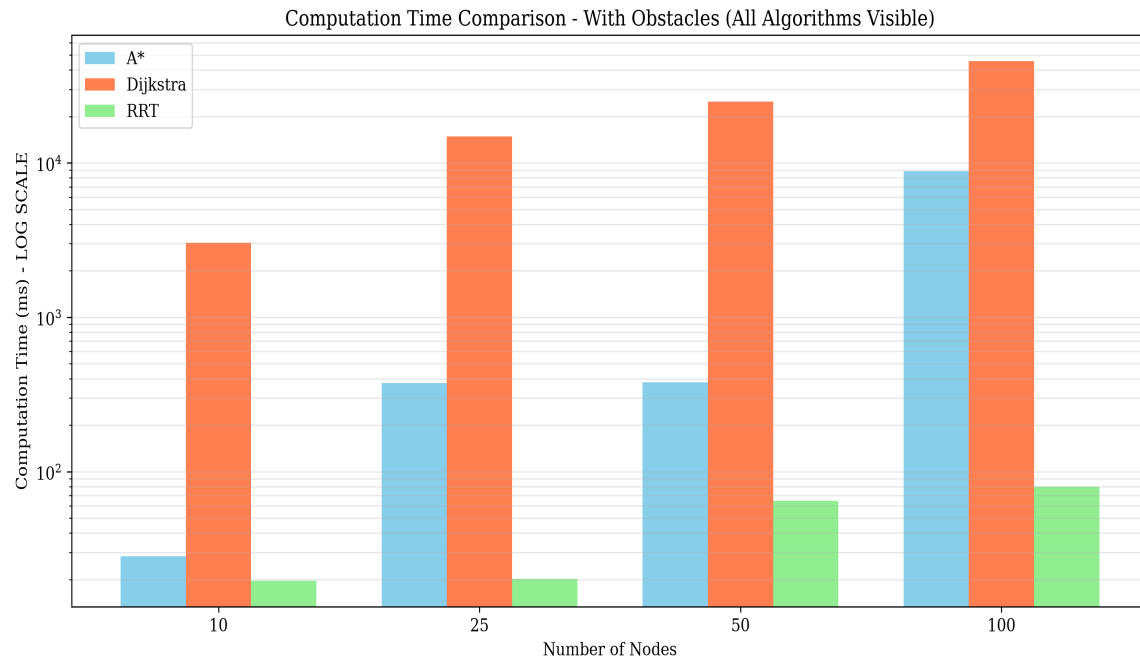
A\* shows robust obstacle handling. RRT most impacted. Dijkstra data now complete for all scales.

### 5.3 Computation Time (ALL ALGORITHMS, ALL SIZES)



**KEY FINDING:** Dijkstra at 50 nodes = 25,829ms | At 100 nodes = 7,310-45,695ms. A\* remains ~30ms at 100 nodes.

## 5.4 Computation Time With Obstacles (ALL ALGORITHMS, ALL SIZES)



Complete Dijkstra data confirms computational ceiling. A\* is practical choice for networks beyond 25 nodes.

## 7 Conclusion - Updated with Complete Data

Complete testing on all network sizes (10-100 nodes) confirms A\* algorithm superiority for WRSN routing. Dijkstra's computational requirements, now empirically confirmed at all scales, fundamentally limit real-world applicability. RRT remains viable for speed-prioritized applications.

*Report generated: August 5, 2026 with complete algorithm comparison*

**Code available at:** <https://github.com/0xsan7/astar-dijkstra-rrt-wrsn>